WASM

Assembler/Disassembler
For Wang Programmable Calculators
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INTRODUCTION

‘wasm’ is an Assembler and Disassembler for Wang 600/700 Programmable Calculators. It recognizes, but does not require, certain file type suffixes. Any suffix of the form “.w6x” (where ‘x’ is any letter) indicates Wang 600 code. Similarly, any suffix like “.w7x” indicates Wang 700 code. A suffix with the last letter ‘t’ is recognized as a tape image, which has additional codes to handle the end of the image.

ASSEMBLER

The assembler converts opcodes (and operands where appropriate) into instruction codes. Comments begin with a semicolon character and extend for the rest of the current line.

The special pseudo-op “ENTER” is used to make entering numbers easier. The operand for “ENTER” is a string of decimal digits with optional ‘E’, ‘-’, and ‘.’ characters. This string represents the sequence of keystrokes to enter the number, and NOT a standard number. In many cases the string representation will be the same as the number, but it need not be. For example, “1.1-”, “-1.1”, “1.-1” all represent the number -1.1.

Errors in assembly are noted by a letter in the first column of the listing output. The lines containing errors are also sent to stderr in the case that no listing is being printed to stdout. The error letters and their meanings are in the “ASSEMBLER ERRORS” section.

DISASSEMBLER

The disassembler may be used to convert program or tape images into source code, allowing existing programs saved from a Wang Programmable Calculator Simulator to be turned into source code, or simply viewed (listed) in a symbolic form.

The disassembler looks for sequences of number entry commands and combines those into a single “ENTER” pseudo-op. It also converts ALPHA text sequences into more-natural strings.

COMMAND LINE

The following command line options are used to control the behavior of ‘wasm’:

600

Use Wang 600 instruction codes and mnemonics. Also allows 601, 602, 611, 612, 607, or 606 to specify the default output device.

700

Use Wang 700 instruction codes and mnemonics. Also allows 701, 702, 711, 712, 707, or 706 to specify the default output device.

docs

Do not assemble/disassemble, only print a list of codes and mnemonics for the specified machine type.

tape

Treat image file (input or output) as a cassette tape image, adding extra codes to mark the end of the image. Each End Program (END) code will appears as two successive END codes, and the end of a data image will have END followed by 15-15.

asm=*file*

Assemble *file* using the (implied or specified) machine type mnemonics and syntax. By default, a listing is produced on stdout and the object code is written to a file named “a.out”.

nolst

Do not produce any assembly listing.

list=*file*

Produce an assembly listing written to *file*, instead of sending to stdout.

out=*file*

Write assembled object code (or tape image) to *file*, instead of “a.out”. Or write disassembly output to *file* instead of stdout.

dis=*file*

Disassemble the program image *file* and send to stdout. By default, this produces a listing-style output (containing step numbers and object code).

raw

The disassembly output is source code, not listing. This code should be ready for re-assembly.

Note that it does not make sense to specify both “dis=” and “asm=” in the same command. If both are specified, “asm=” will be ignored.

OPERAND FORMATS

ALPHA

The “ALPHA” mnemonic, which represents the “α” key on the Wang 600 or the “WRITE ALPHA” key on the Wang 700, takes an operand that may be a single specific key (mnemonic) or a numeric code (e.g. 00-00), or a string enclosed in double quotes. In the case of a string, the ALPHA command is coded as a variable-length command that is automatically terminated by the appropriate code based on machine type. Within the string, shift codes are handle automatically. The following special codes (escapes) are used to represent special characters:

“\r”	RETURN-INDEX (carriage-return line-feed)
“\b”	BACKSPACE
“\t”	TAB
“\i”	INDEX (line-feed)
“\v”	REVERSE INDEX
“\h”	“½”
“\q”	“¼”
“\c”	“¢”
“\n”	DC2 a.k.a. PUNCH-ON
“\f”	DC4 a.k.a. PUNCH-OFF

Not all output devices support all these characters.

REGISTER DIRECT

Register direct instructions use the decimal value of the register index, the same number that would be used for indirect register access (not the coded value placed in the program step). For the Wang 700, this includes automatic handling of the “register + 100” cases.

PRINT / WRITE

The PRINT (Wang 600) or WRITE (Wang 700) instruction is followed by a code that indicates the format used to print the number. The Wang 600 uses a format that specifies a letter “tag” and the number of decimal places separated by a slash, for example “X/02”. The Wang 700 uses a format that specifies the number of places to blank-pad in the whole-number portion and the number of decimal places, represented as a standard program code, for example “05-02”.

MARK / SEARCH

Instructions that use a statement label may specify the label as either a standard code (“00-00”) or by the key mnemonic (“E0”). The disassembler favors the key mnemonic.

ASSEMBLER ERRORS

The following error letters are used:

V	Invalid characters in ENTER operand.
O	Invalid opcode mnemonic.
S	Syntax error (missing operand).
P	Operand error.
R	Register value error.
F	Printer format error.
I	I/O operand error.

ASSEMBLER PSEUDO-OPS

The following pseudo-ops are recognized:

.REG [*label***] [“***string***” | ***value***]**

Define register space inside the program. The program is padded with STOP instructions to the next register location. If *label* is specified, it will be assigned the register number corresponding to the location in the program. It may be used by prefixing with “&” in register-direct instructions. If the machine stores two registers in a program block (as the 700 does), then a second label and second value may be specified, separated by a comma. If *label* is “-” then no label is generated (but the value field is still recognized). The *string* is a hexadecimal string of 16 digits, representing the codes to be stored in each digit location in the register. The *value* is a floating point value in standard notation, for example “1” or “1.2e6”.

.OUT *device*

Specify the output device type that applies to subsequent ALPHA text instructions. This determines the character set available and how strings are converted to Wang character codes. The device value must match a valid output device for the Wang machine family, for example the model 601 is the Output Writer for the Wang 600, while 701 is the Output Writer for the Wang 700. The following models are supported:

601/701	Basic Output Writer
602/702	Plotting Output Writer
611/711	Input/Output Writer
612/712	Flatbed Plotter
607/707	Teletype

.INCLUDE *file*

Include the specified file at the current location in the program. There is currently no search path, so file must explicitly name the file relative to the current directory when running “wasm”.

EXAMPLES

This example uses the “.REG” psuedo-op to create a register within a program that contains a pattern for a blank display on the Wang 600:

```
...
      RECALL &blank
      ALPHA STOP
      ALPHA STOP
...
      .REG blank “FFFFFFFFFFFFF00A”
      END
```

Note that, due to wasm being a single-pass assembler, the listing output shows a dummy value in the RECALL instruction even though the final program file contains the actual register number code.

There is currently no way to tell the disassembler about sections of a file that contain register data.

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